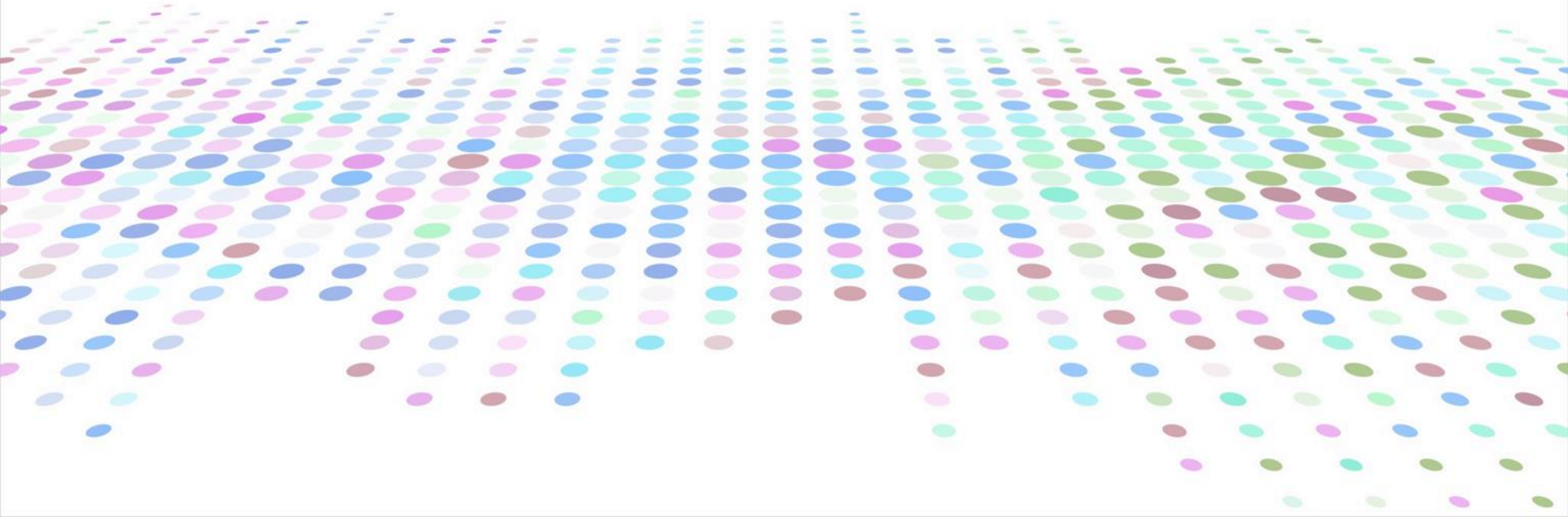
Ligaments & Cartilage Injuries

Irvind Sihota (i.sihota@londonmet.ac.uk)



Ligaments & Cartilage Injuries

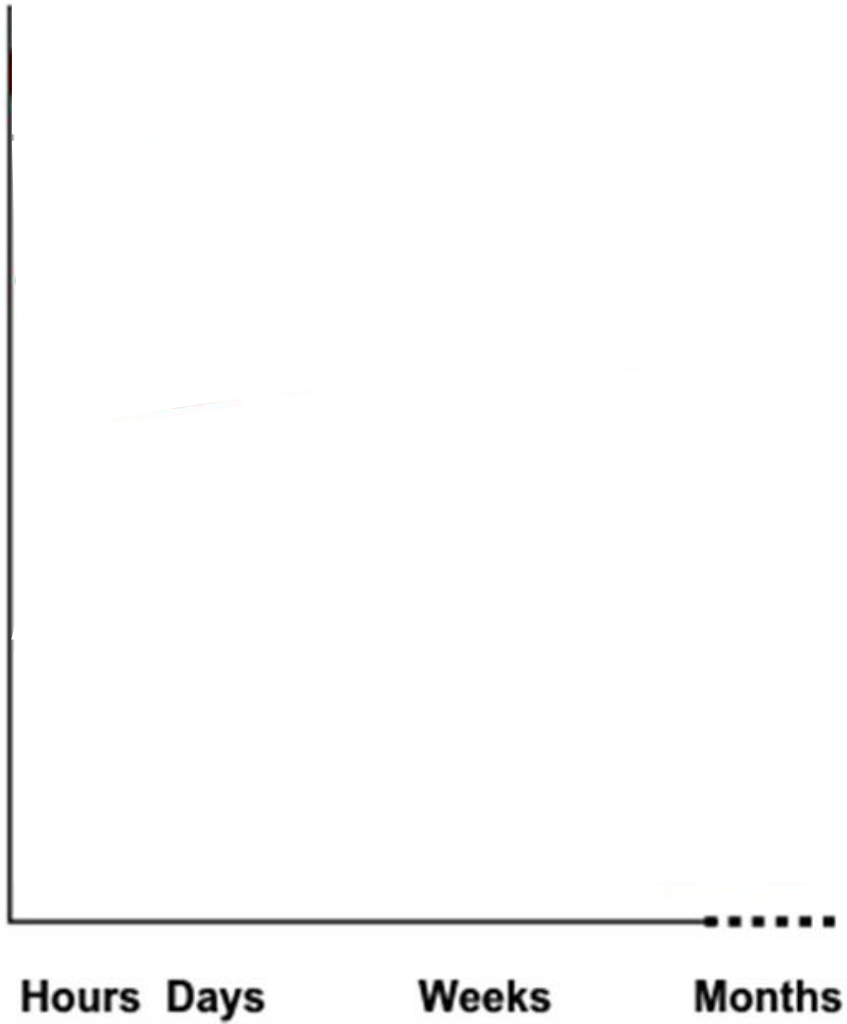
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Irvind Sihota
(i.sihota@londonmet.ac.uk)

Ligaments & Cartilage Injury

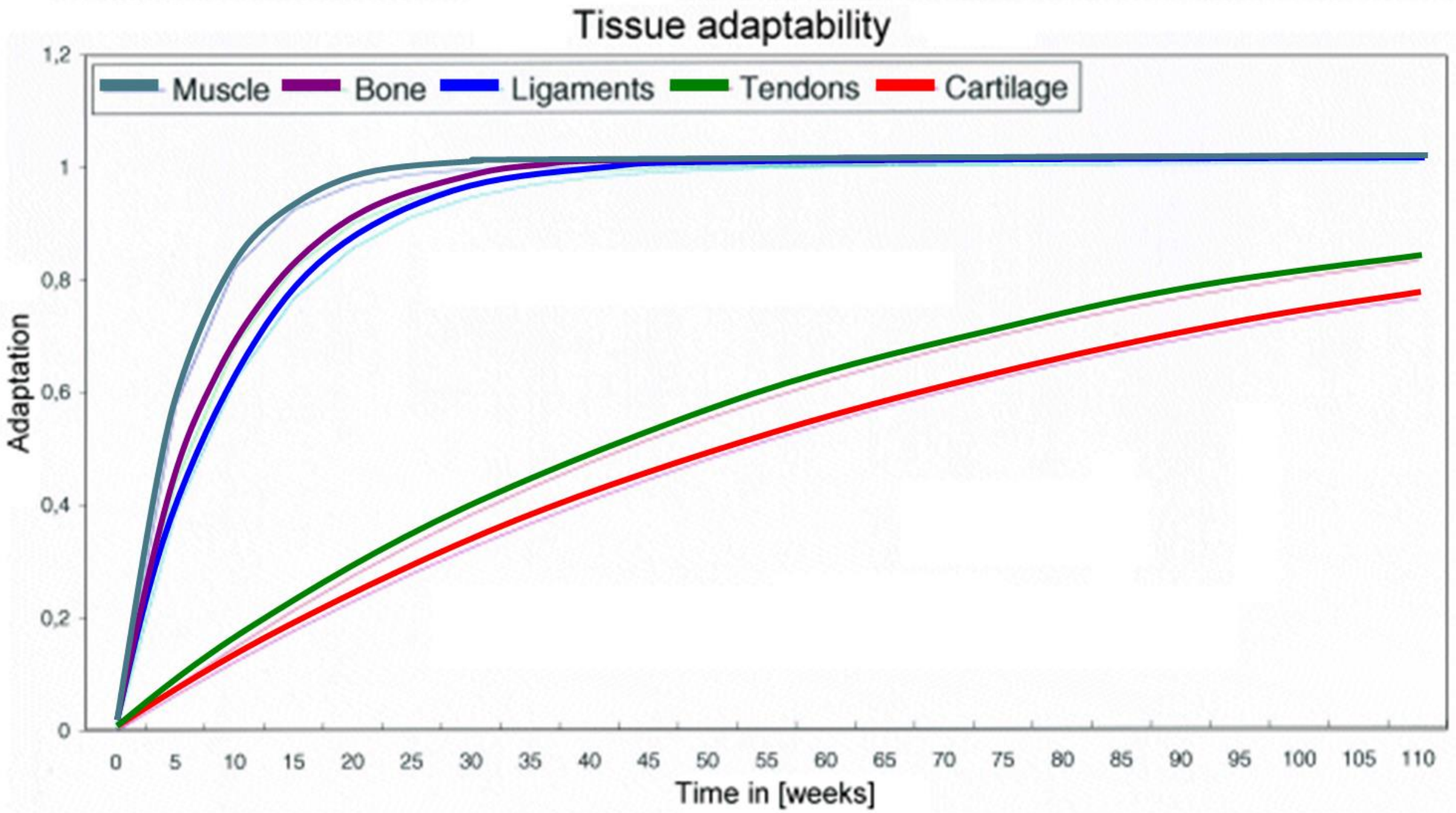
- Discuss the incidence and mechanisms of ligament injuries
- Discuss evidence-based management approaches for ligamentous injuries
- Design a management plan for case studies
- Introduction to Cartilage Injuries

Healing Process



Ligament Healing Video

https://www.youtube.com/watch?v=koWYTCP-A8A


Tissue adaptability
Muscle
Bone
Ligaments
Tendons
Cartilage
Adaptation
1,2
1
0,8
0,6
0,4
0,2
0
0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100 105 110
Time in [weeks]

Lateral Ankle Sprains most common Injury (52.7 per 10,000 in Emergency Departments)

Most common causes for PCL injuries are road traffic accidents and athletics.

70% of general population have reported an ankle injury

70% of all ACL injuries are non-contact (30% contact injuries).

https://cks.nice.org.uk/topics/sprains-strains/background-information/prevalence/



Function

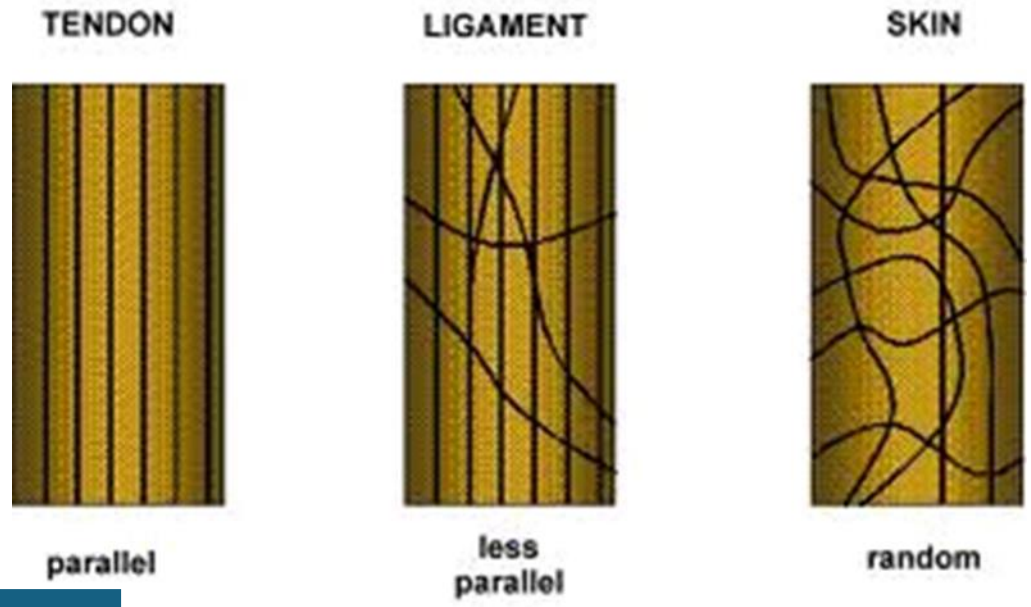
- Connect bone to bone
- Give mechanical stability to the joints
- Static non-contractile constraints
- Passive stability



Structure

Component	Ligament	Tendon
Cellular material - Fibroblasts	20%	20%
Extracellular matrix	80%	80%

Extracellular matrix: water (70%) and solids (30%)



Component	Ligament	Tendon
SOLIDS		
Collagen	**70-80%**	**Slightly higher**
- Type I	**90%**	**95-99%**
- Type III	**10%**	**1-5%**
- Ground substance **(Proteoglycans, glycoproteins, plasma proteins)**	**20-30%**	**Slightly less**
- Elastin (small amounts)		

Factors affecting properties of ligament

- Maturation and aging
- Pregnancy
- Mobilisation
- Immobilisation
- Diabetes
- Steroids
- NSAIDs
- Grafts



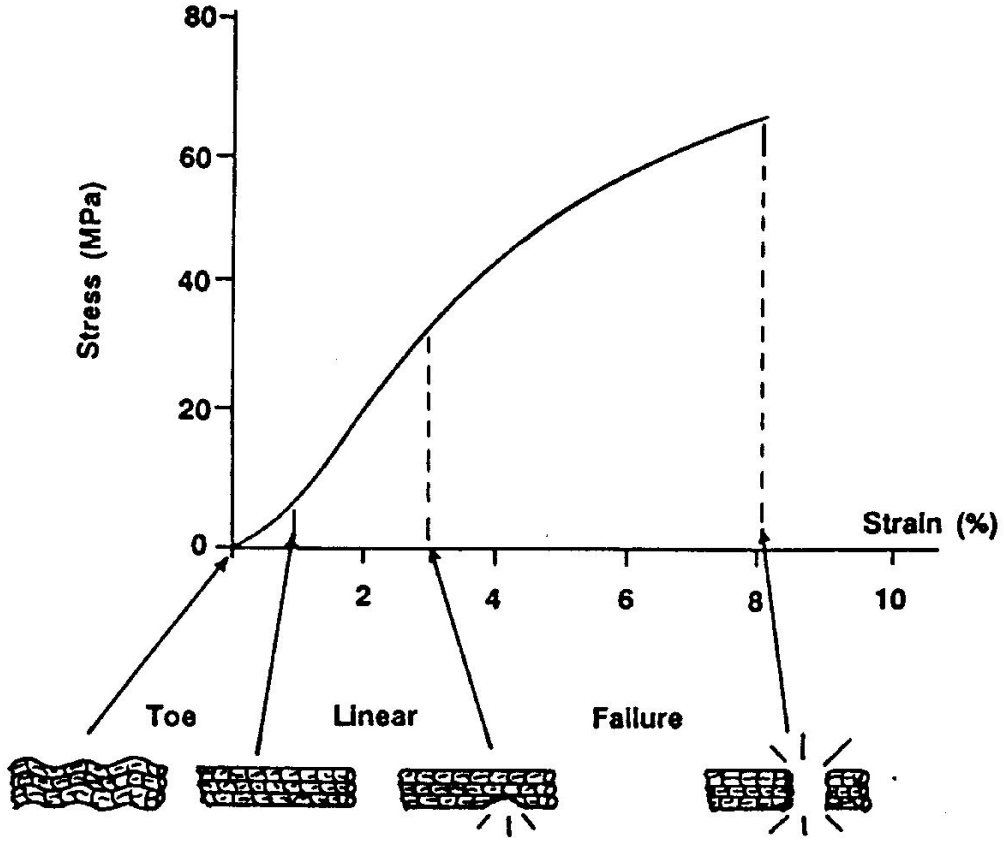
Pathophysiology

Acute injuries

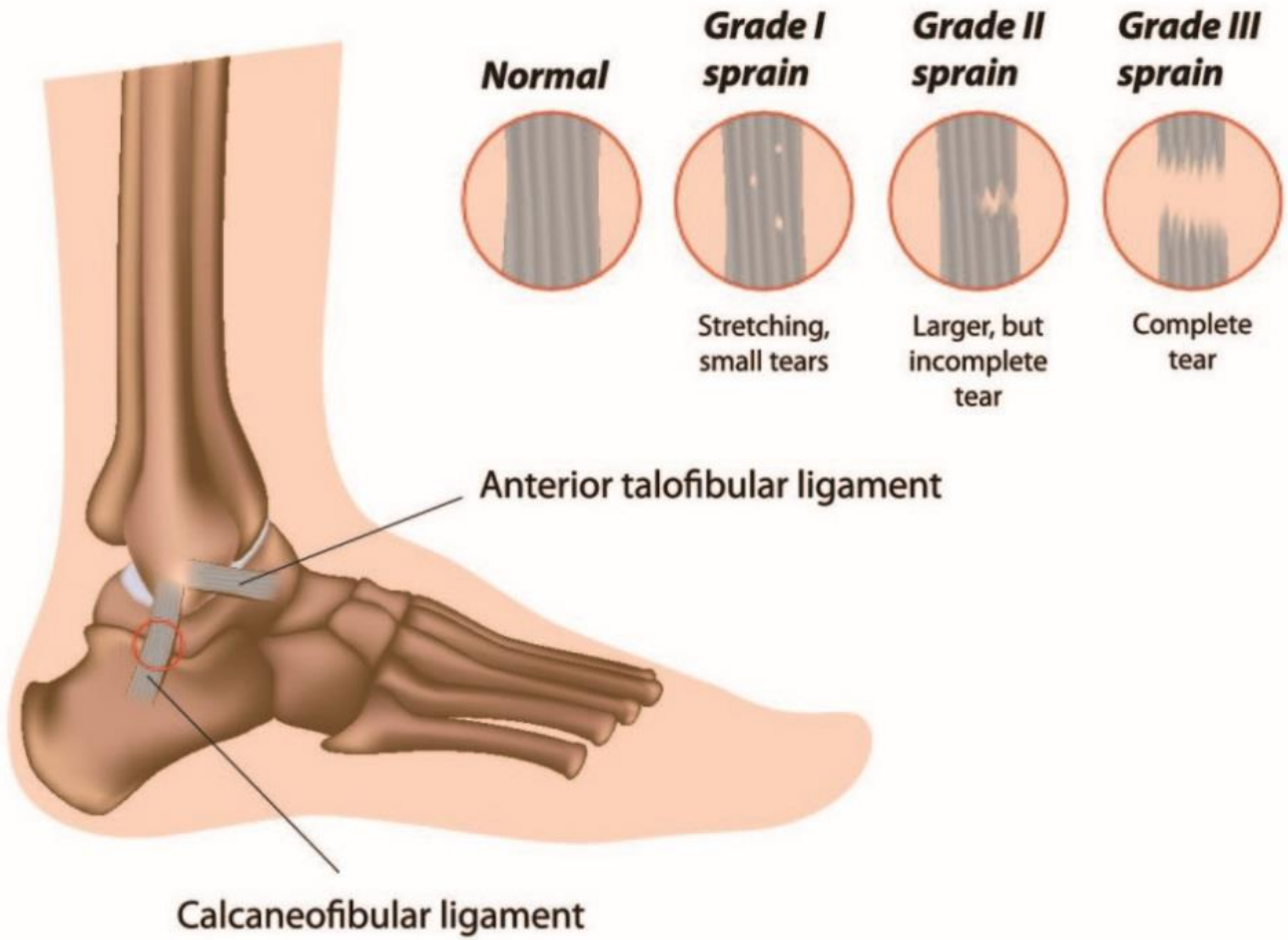
- Why did they occur on a microscopic level
- Need to know structure, function and composition
- Repair process

Chronic injuries

- Process undergone to lead to the injury
- Cellular level
- Repair process



- Elastin largely effects this graph, with more elastin allowing the material to return to its original shape following greater stress being applied
- Think about the implications of the structure becoming less elastic? What happens to the graph? Why might it become less elastic?



Sprain

Ligaments = Sprain

West Point Sprain Grading System				
	Tearing	Swelling	Joint Instability	Weight Bearing
Grade I	microscopic	minimal	none	fully / partial
Grade II	partial	moderate / severe	mild / moderate	unable
Grade III	complete rupture	severe	moderate / severe	unable

Staging initially established for different treatment plans, but now regardless of staging all complete the same treatment plan--functional rehabilitation.

Haemarthrosis and Effusion

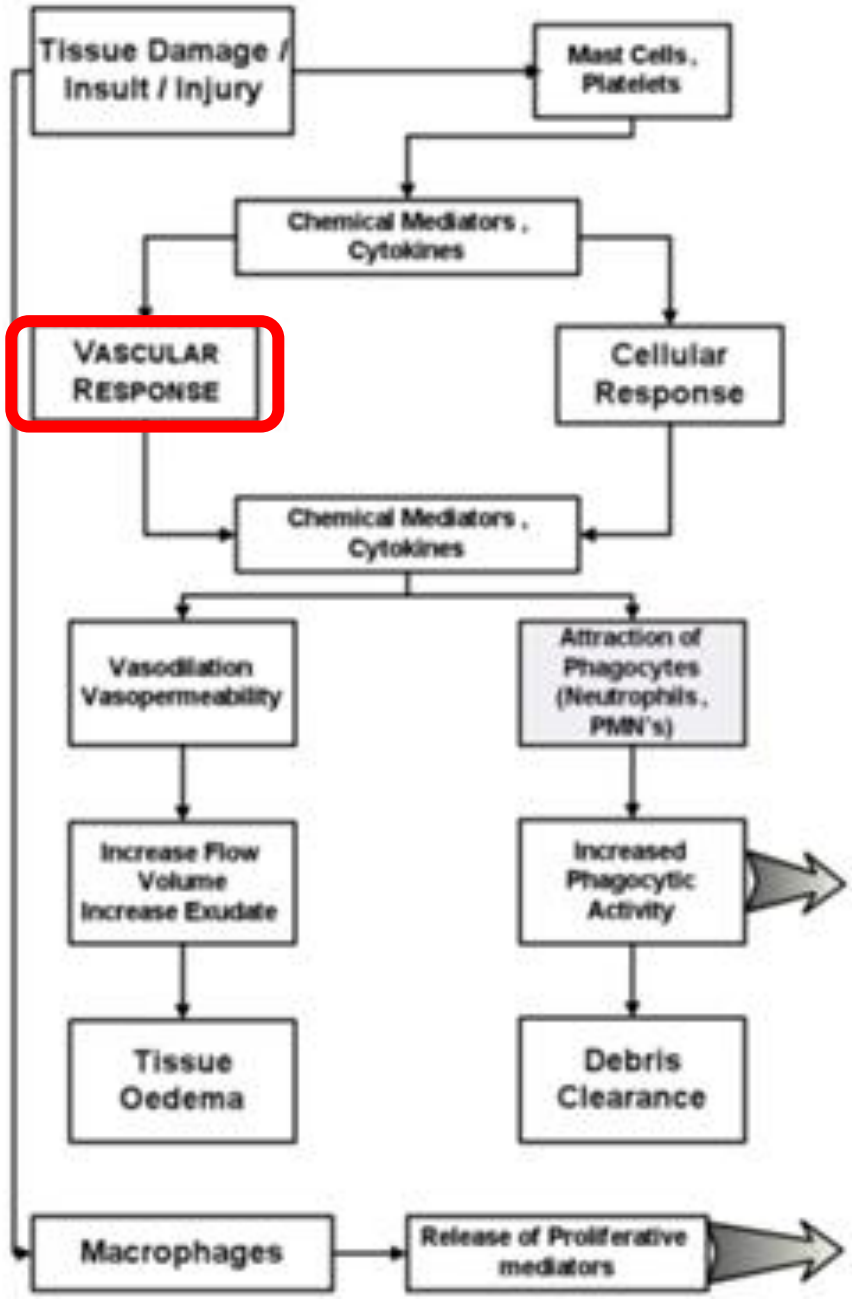
Haemarthrosis	Effusion
Swelling seen immediately after the injury : 15-20 minutes or sooner	Swelling comes after >6-12 hours of injury
Indicates cruciate or peripheral meniscal lesion in the absence of a fracture	Indicates a tear of the meniscal body or synovial injury / irritation
Patella tap positive	Patella tap positive

Healing

- Inflammation (1 – 3 days following)
  - Platelets > fibrin clot > growth factors > inflammatory cells infiltrate > fibroblastic scar forms
  - Maximal reaction 1-3 days following
  - Occurrences:
    - Cell death
    - Release of chemicals / inflammatory mediators
    - Broken blood vessels

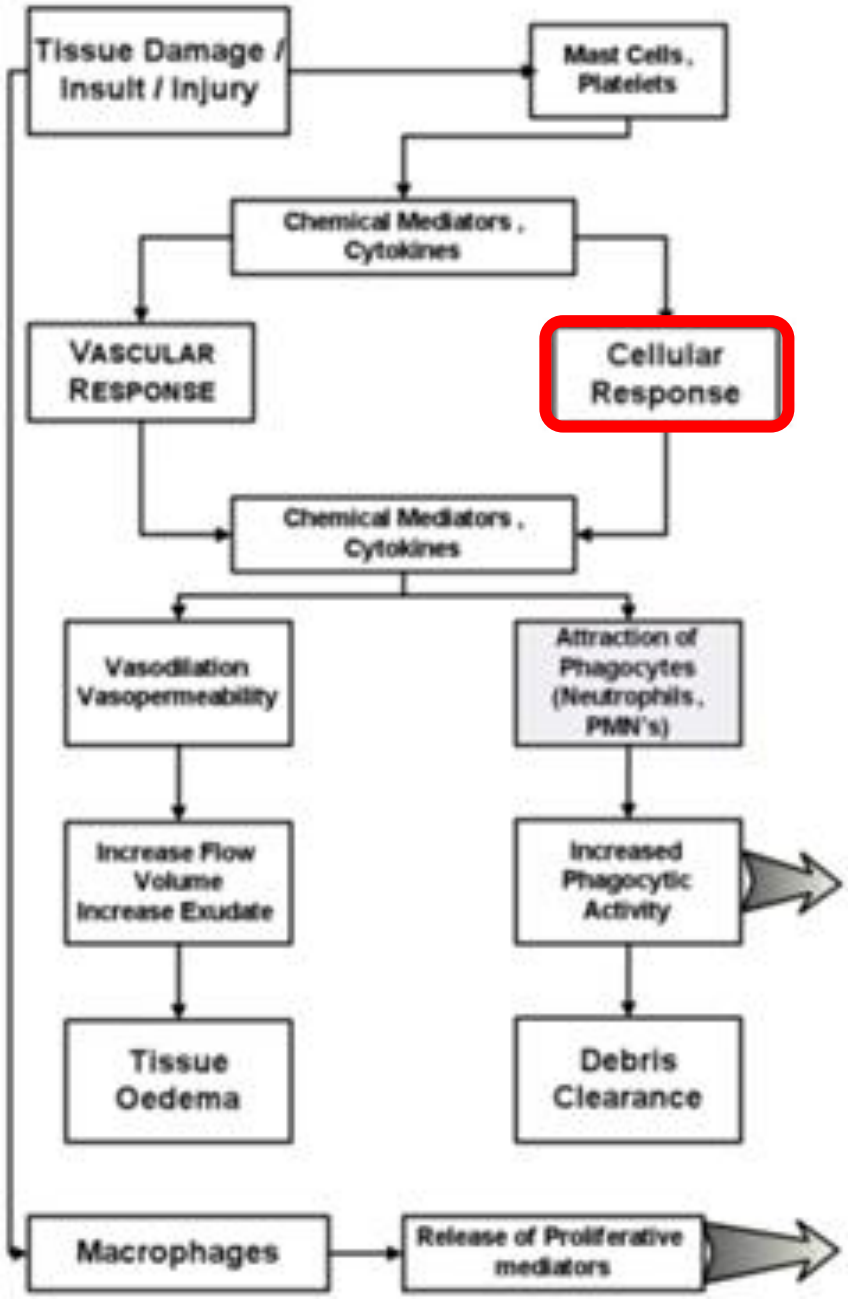
Healing

- Proliferation
  - Scar matrix forms
  - Collagen is smaller in diameter than normal
  - More haphazard than normal
- Remodelling
  - Becomes less viscous
  - More dense and organised
  - Never the same as ‘normal’ ligament



Vascular events

- **VASODILATION** follows brief vasoconstriction
- Increase volume of platelets and white blood cells
- Vasodilation caused by chemical release (Histamine & Serotonin)
- **VASOPERMEABILITY**, also mediated by chemical release = increase leakage
- Increase presence of plasma proteins which causes attraction and retention of fluid = swelling



Proliferation

Three phases of tissue restoration;

1. **R**ESOLUTION – Dead cells & cellular debris are removed
2. **R**EGENERATION – Damaged tissue is replaced by cells of same type
3. **R**EPAIR – Original tissue is replaced with scar tissue

Two processes of repair;

1. FIBROBLAST FORMATION
2. ANGIOGENESIS

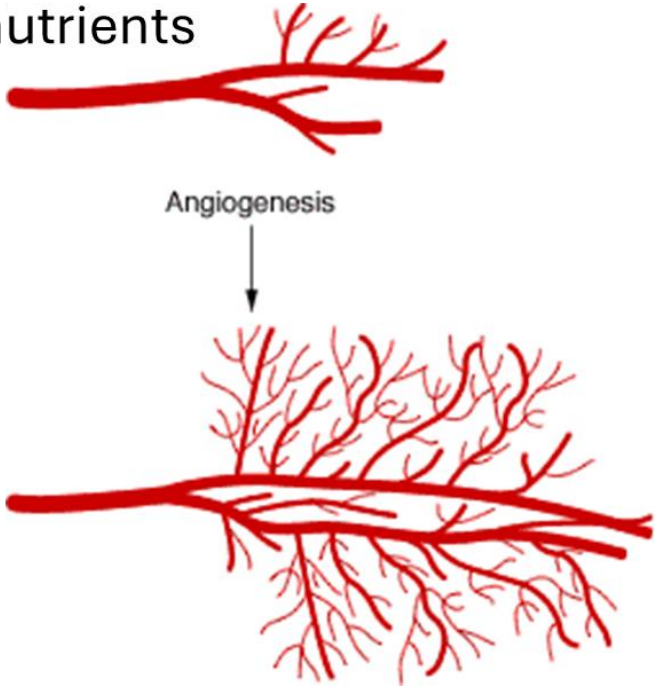
Fibroblast formation

- Fibroblast = “A cell that manufactures and maintains connective tissue”
- Migrate to the area and produce new tissue
- Initially produce type III collagen – granular tissue



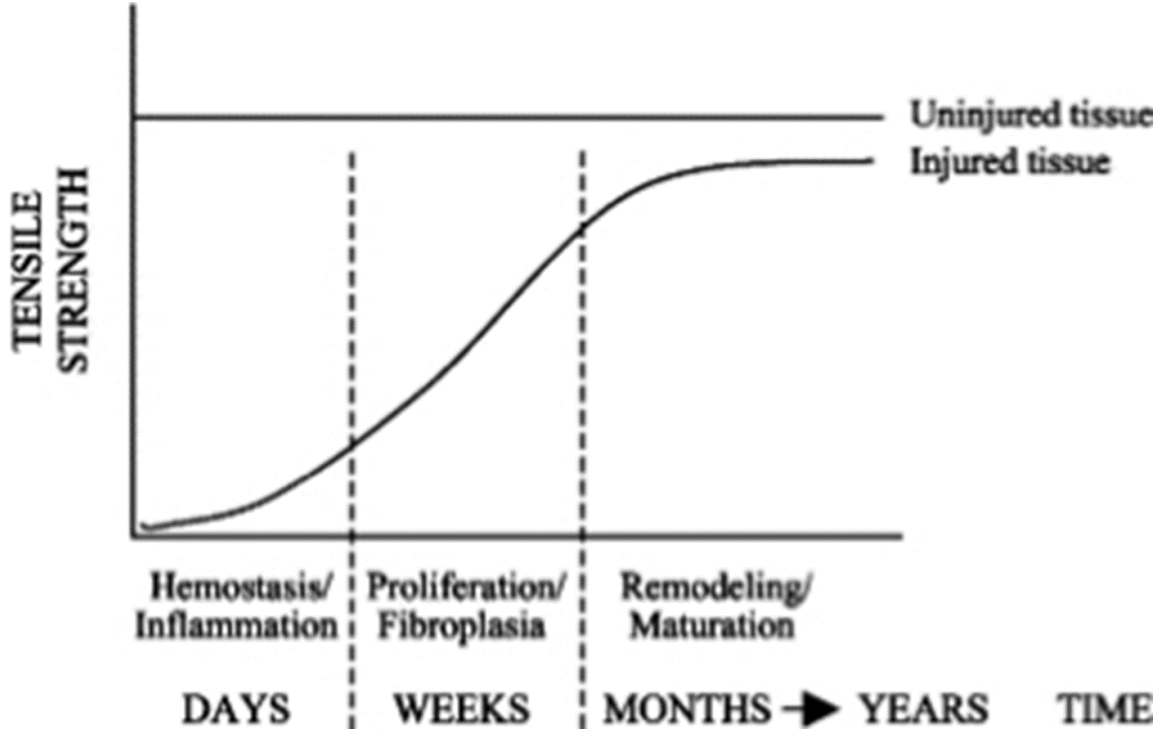
Angiogenesis

- Angio – Latin for ‘blood vessel’ / Genesis – ‘to create’
- Capillaries bud in area to provide cells with blood which provide oxygen and nutrients

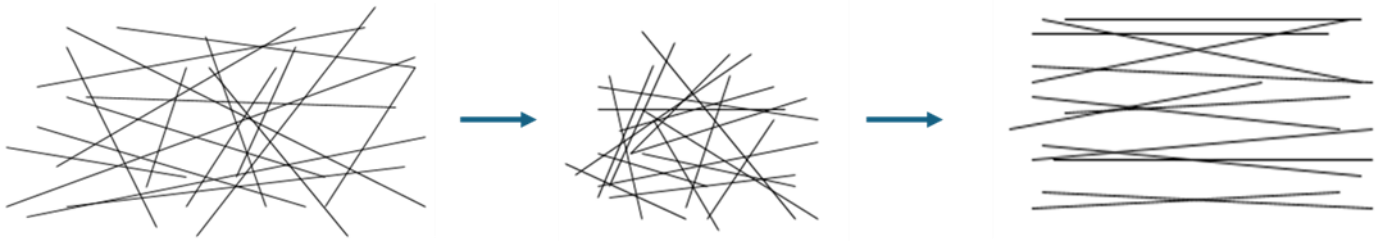


Remodelling

- Refinement of the collagen -> up to 12 months post injury
- Initial collagen deposition results in weak fibrils with random orientation.
- Remodelling and maturity = collagen becomes shorter and more orientated.

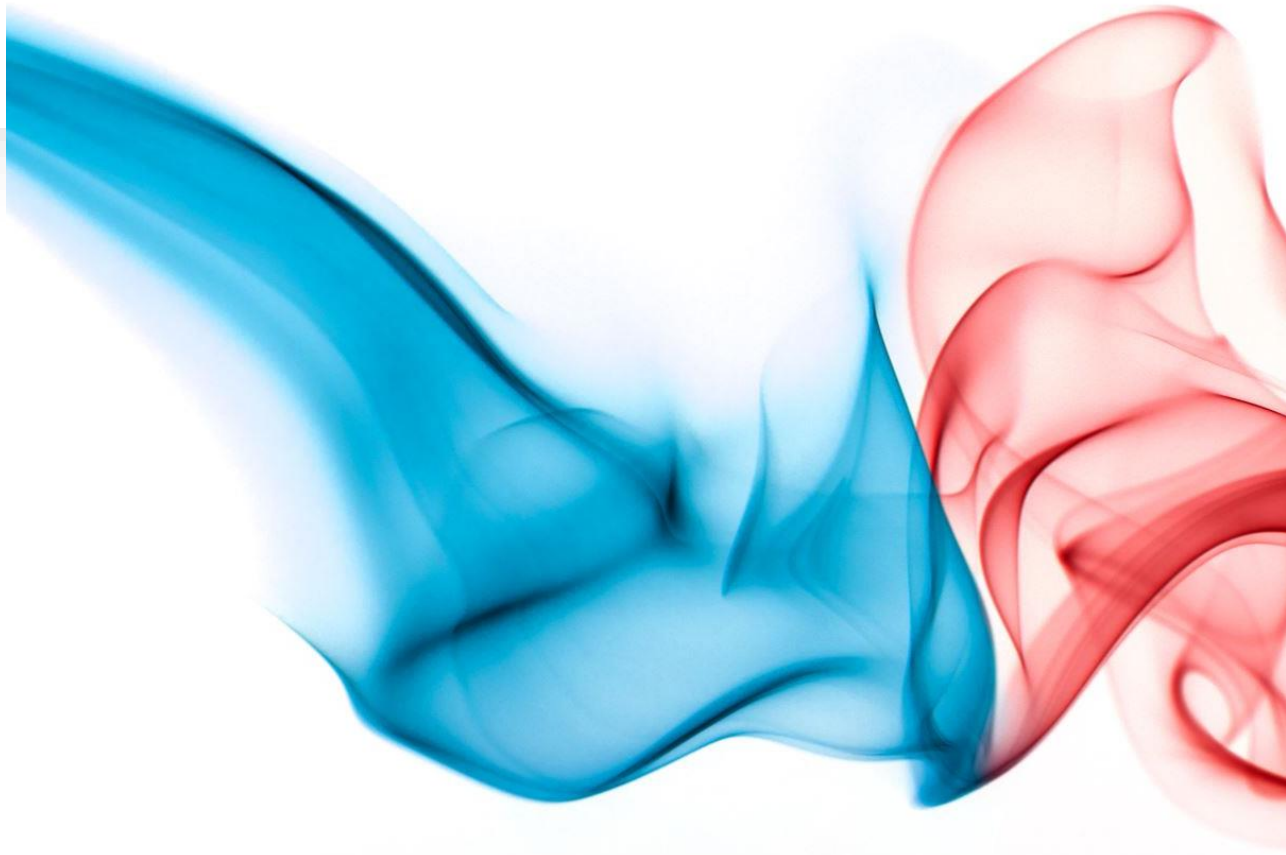


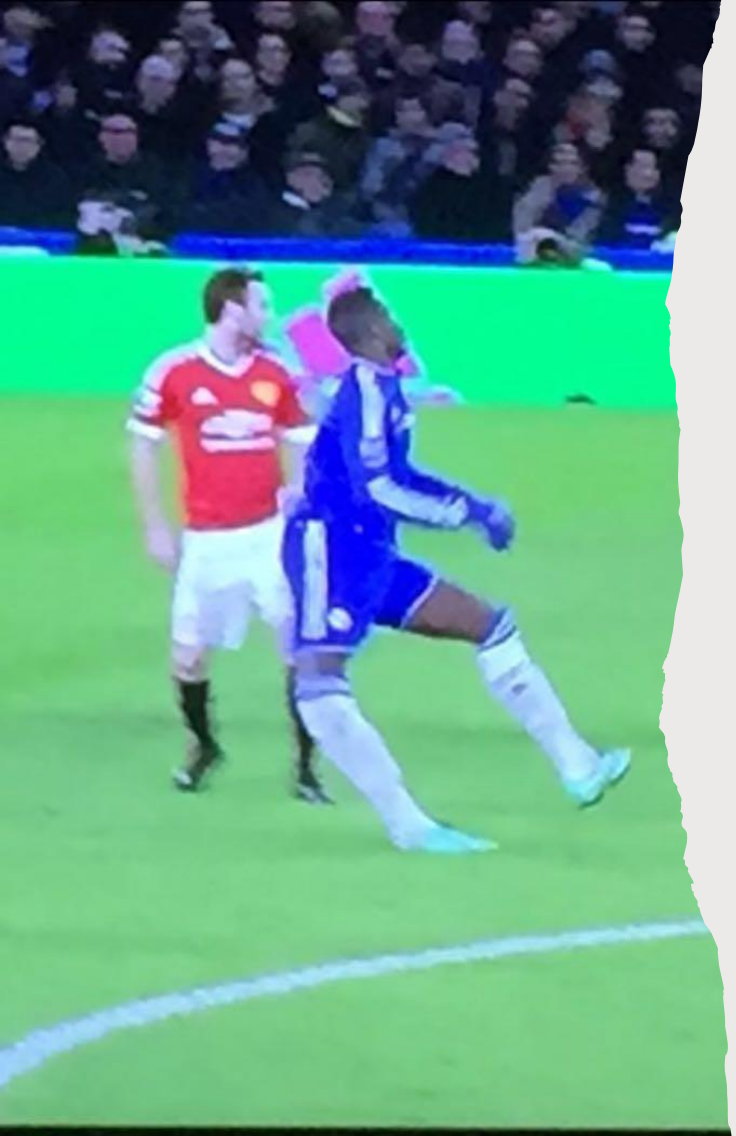
Type III collagen → Type I collagen



Factors that affect healing

- Age
- Diabetes
- Smoking
- High BMI
- Steroids & NSAID's
- Temperature (lower rate when colder)
- Poor blood supply / ischaemia
- Excessive movement





Diagnosis

- Subjective history – Symptoms / Mechanism of Injury / Sounds?
- Observation - swelling
- Change in ROM
- Inhibitory strength loss
- Ligament stress tests
- Palpation
- Specific special tests e.g. anterior drawer
- MRI



83 40

Ligament Injury

- Management
- Partial tear may or may not require surgery
- Complete tears, especially in younger
- athletes - may require reconstruction

- Conservative
- Used in overall low activity level patients
- If the overall stability of the knee is good
- Involves treatment program of:
  - muscle strengthening
  - use of a brace to provide stability

Ligament Injury

- Management
- Operative treatment can involve:
- Arthroscopic
- Open surgery
- Combination
- Procedures can involve:
- Strip of tendon, taken from the patient's knee
- patellar tendon
- hamstring tendon (semitendinosus)
- passed through joint and secured to femur & tibia

Table 1
Classification of Ankle Sprains

Severity	Physical Examination Findings	Impairment	Pathophysiology	Typical Treatment*
Grade 1	Minimal tenderness and swelling	Minimal	Microscopic tearing of collagen fibers	Weight bearing as tolerated No splinting/casting Isometric exercises Full range-of-motion and stretching/strengthening exercises as tolerated
Grade 2	Moderate tenderness and swelling Decreased range of motion Possible instability	Moderate	Complete tears of some but not all collagen fibers in the ligament	Immobilization with air splint Physical therapy with range-of-motion and stretching/strengthening exercises
Grade 3	Significant swelling and tenderness Instability	Severe	Complete tear/rupture of ligament	Immobilization Physical therapy similar to that for grade 2 sprains but over a longer period Possible surgical reconstruction

*Patients must receive treatment that is tailored to their individual needs. This table outlines common treatment protocols.

Reprinted with permission from Bernstein J (ed): Musculoskeletal Medicine, Rosemont, IL; American Academy of Orthopaedic Surgeons, 2003, p.242

Management of Ligaments Injuries

Sprains

Table 1
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Reprinted with permission from Bernstein J (ed): Musculoskeletal Medicine,

Phase 1 – Immobilisation / Protection – Day 1 - 4

- Elbow crutches supplied to offload – hinged brace not required (advised by Orthopaedic)
- Given icing strategies
- Advised on analgesics
- Active assisted knee flexion and extension
- Quad isometrics

Phase 2 – Early motion - Day 4 - 14

- Removal of elbow crutches
- Gait re-education
- Static cycling
- Seated double leg leg-press
- Supported upper body / trunk conditioning
- Hydrotherapy



Phase 3 – Strengthening – Day 14 - 30

- Continued static cycling – increasing resistance
- Continued upper body / trunk conditioning
- Sagittal plane resistance training
- Straight line running
- Progressive proprioception drills – Double leg > single leg





Proprioception

https://www.youtube.com/watch?v=ZrDgUwrUPSU

Phase 4 – Functional activity – Day 30 - 40

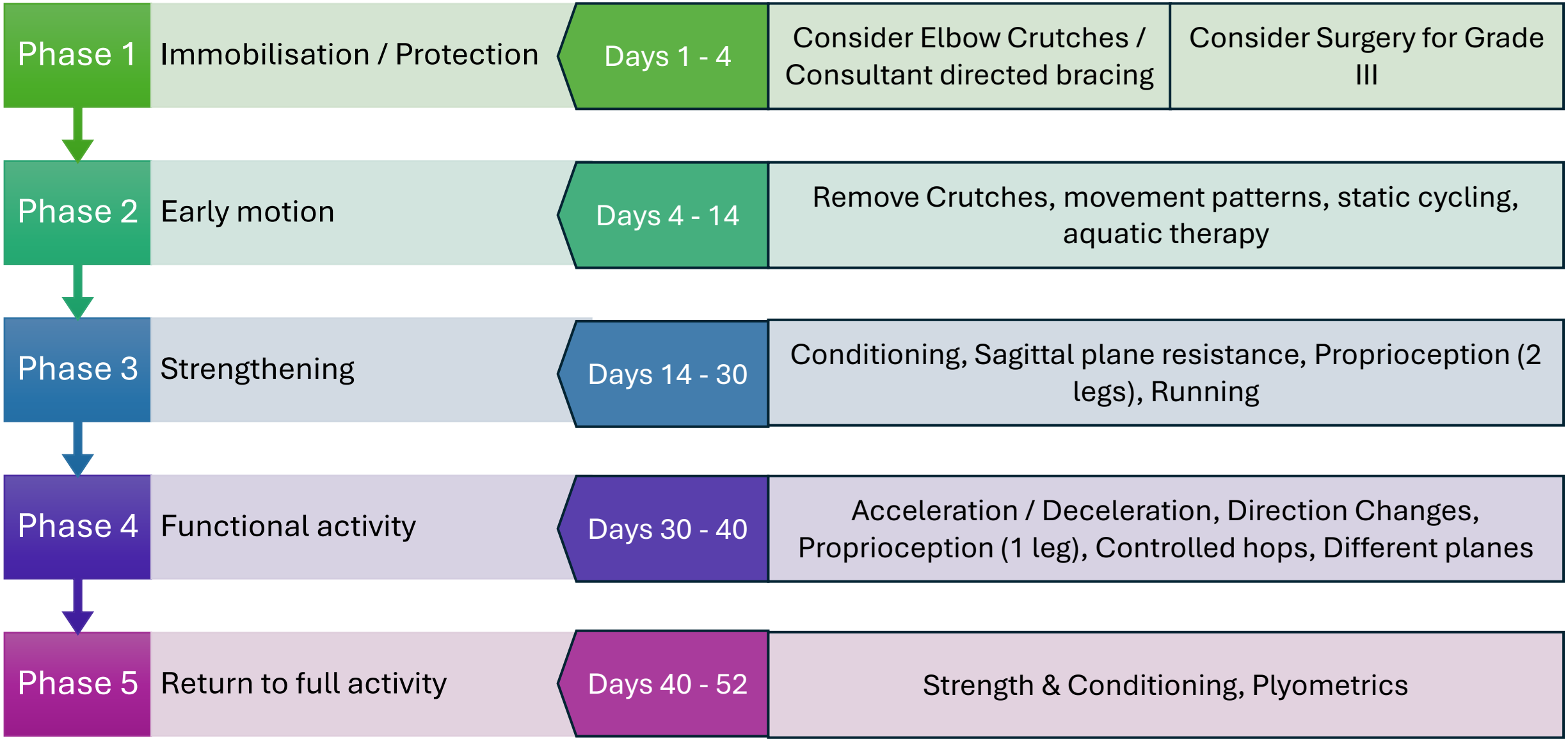
- Acceleration / Deceleration running
- Controlled changes of direction
- Progressive single leg proprioception
- Controlled hops
- Sports specifics in different planes



Phase 5 – Return to full function – Day 40 - 52

- Return to lower limb strength and conditioning
- Cutting and turning
- Plyometrics
- Return to non-contact sparring

Phases of the rehabilitation


Phase 1
Immobilisation / Protection
Days 1 - 4
Consider Elbow Crutches / Consultant directed bracing
Consider Surgery for Grade III
Phase 2
Early motion
Days 4 - 14
Remove Crutches, movement patterns, static cycling, aquatic therapy
Phase 3
Strengthening
Days 14 - 30
Conditioning, Sagittal plane resistance, Proprioception (2 legs), Running
Phase 4
Functional activity
Days 30 - 40
Acceleration / Deceleration, Direction Changes, Proprioception (1 leg), Controlled hops, Different planes
Phase 5
Return to full activity
Days 40 - 52
Strength & Conditioning, Plyometrics

Immobilisation?

- Complete immobilization of the knee for an extended period is generally contraindicated because of the prolonged stiffness, muscle atrophy, and chronic pain that result
- Indications for the use of a knee immobilizer include the acute (or presurgical) management of:
  - Quadriceps rupture
  - Patellar tendon rupture
  - Medial collateral ligament rupture
  - Patellar fracture or dislocation
  - Limited number of other acute traumatic knee injuries

Cartilage

- 3 types of cartilage:
  - **Elastic** – e.g. epiglottis, earlobes
  - **Fibro** – e.g. syndesmosis, pubic symphysis, meniscus, labrum
  - **Hyaline** – at the ends of bones

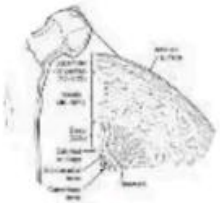
Cartilage

Cartilage acts to:

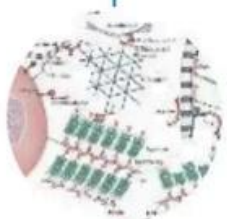
- Provides shock absorption & cushioning in joints
- Reduce friction with joints

Chondrocytes are the cells that form cartilage, and aid endochondral ossification (which is the process of formation and healing of bone)

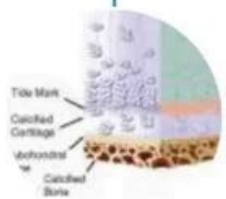
ANATOMY



Components



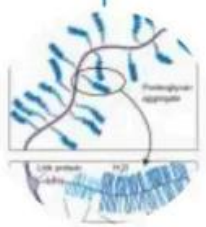
Extracellular Matrix (ECM)



Cellular



Collagen



Proteoglycan



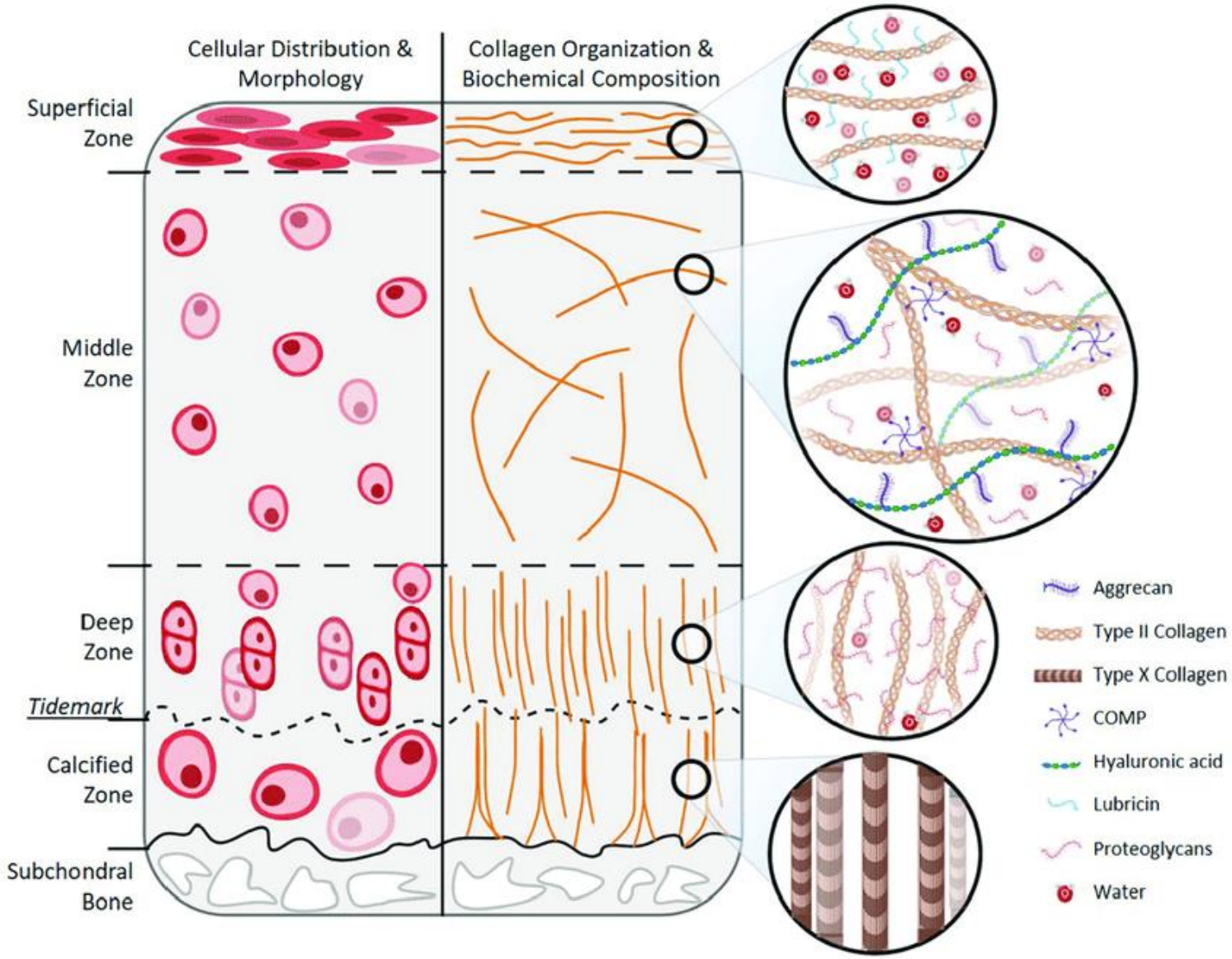
Water



Chondrocyte

Amounts of elastin in dry mass of human tissues.

	Elastin Amount (%)
Large arteries	>50
Yellow ligament	~47
Auricular cartilage	15
Intervertebral disc	1.7, 2
Meniscus	0.6



Thorp, Hallie & Kim, Kyungsook & Kondo, Makoto & Maak, Travis & Grainger, David & Okano, Teruo. (2021). Trends in Articular Cartilage Tissue Engineering: 3D Mesenchymal Stem Cell Sheets as Candidates for Engineered Hyaline-Like Cartilage. Cells. 10. 643. 10.3390/cells10030643.

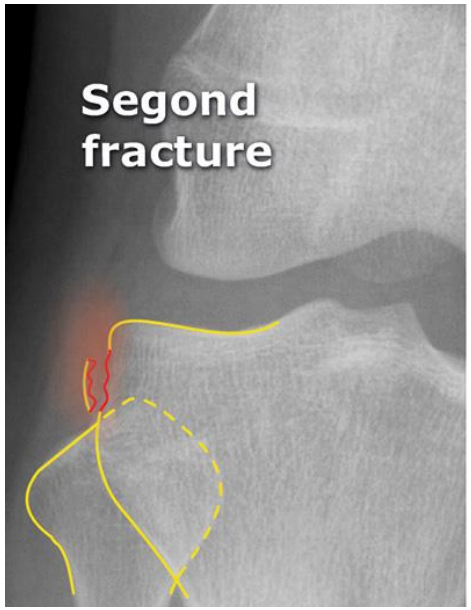
Composition of Hyaline Cartilage

- Composed of:

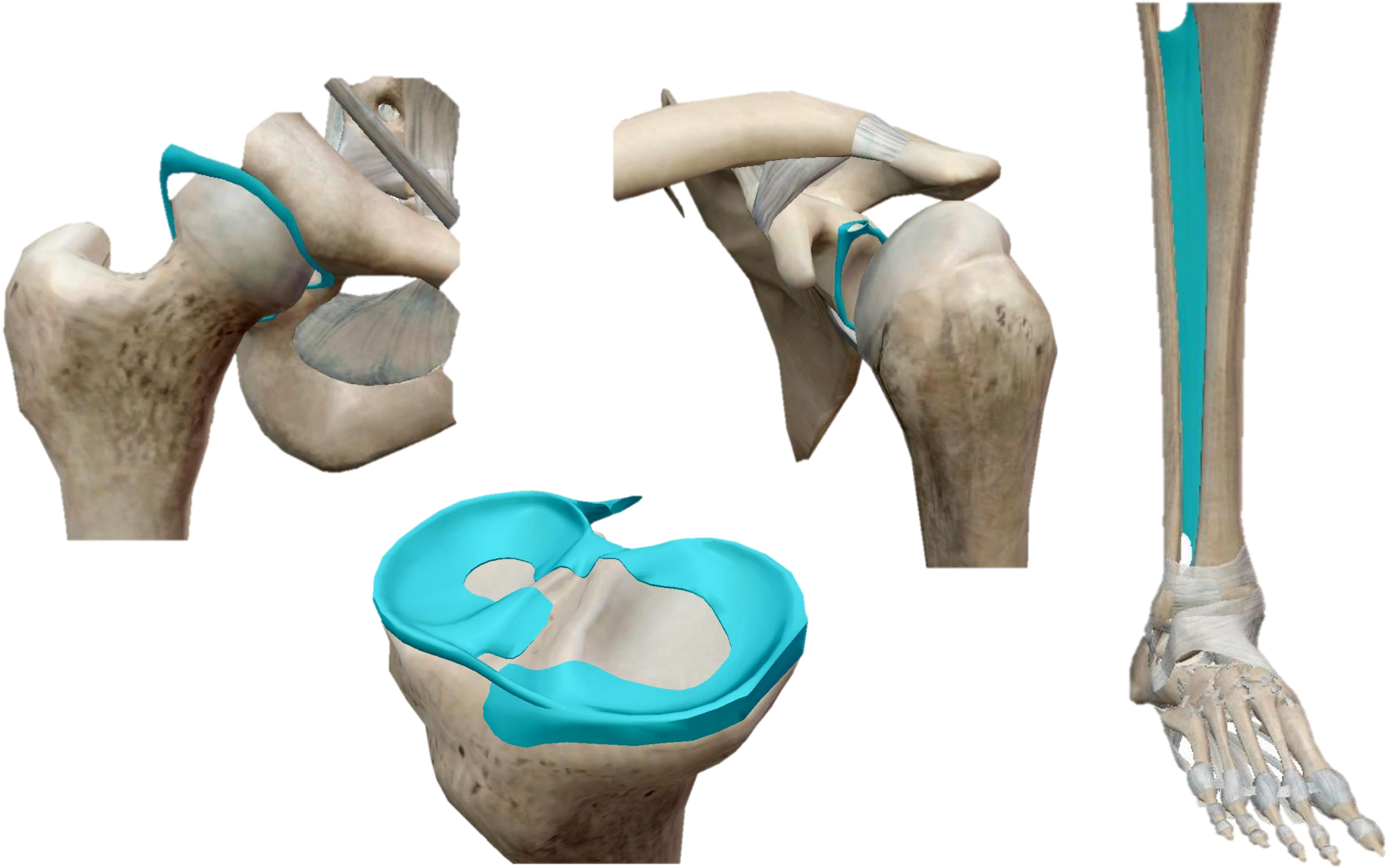
Component	%
Chondrocytes	< 10%
Network of collagen fibres (type 2 collagen)	15-22 %
Concentrated Proteoglycans	4 - 7 %
Water Inorganic salts Other proteins	60-80%

Hyaline Injuries

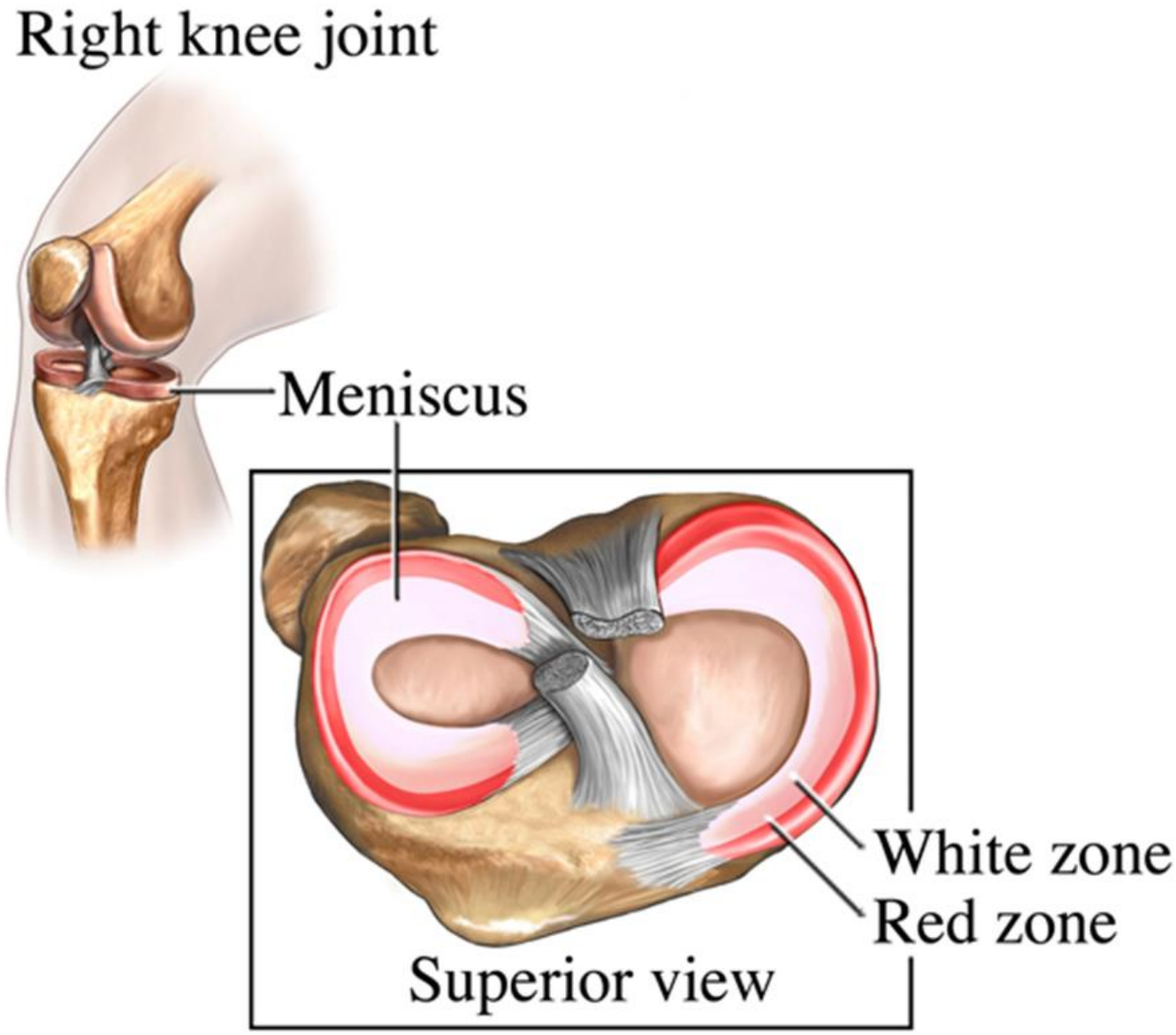
- Degeneration – chronic
  - Loss of structures at a cellular level e.g. OA
- Osteochondral fracture – usually traumatic e.g. Segond # w/ ACL tear
- Osteochondritis Dissecans (OCD) – cell death of underlying bone (due to lack of blood supply) leading to breaking away



Examples of Fibrocartilage



Fibrocartilage Injuries





Fibrocartilage Injuries

Cartilage Meniscus Tears

Types of meniscal tears:

- Parrot Beak Tear
- Horizontal Flap Tear
- Radial Tear
- Vertical Longitudinal Tear
- Horizontal Tear
- Free-edge fraying
- Complex Tear

Injuries

Fibrocartilage e.g. meniscus has similar injuries

- Degeneration
  - Loss of structures at a cellular level
- Tear – usually traumatic
  - Chondral fracture/lesion
  - Flap tear
  - Non-traumatic may be familial or due to OCD (see below)
- Osteochondritis Dissecans – cell death of underlying bone (due to lack of blood supply) leading to breaking away
- Osteochondral fracture – usually traumatic

Aetiology – Degeneration

- Limited capacity for repair as poor blood supply
- **Magnitude of stress**
  - Area over which this occurs – related to biomechanics
  - Body weight
- **Number of sustained peaks of stress**
  - Abnormal number e.g. repetitive heavy impact exercise

- Changes in the proteoglycan matrix leading to reduced integrity
- Deficiencies in other force absorbing structures e.g. Meniscus
- Often the end result is OA or chondral defects

Cartilage Injury Video

https://www.youtube.com/watch?v=mr5JI8Q8dc8&t=161s

Cartilage Surgery Video

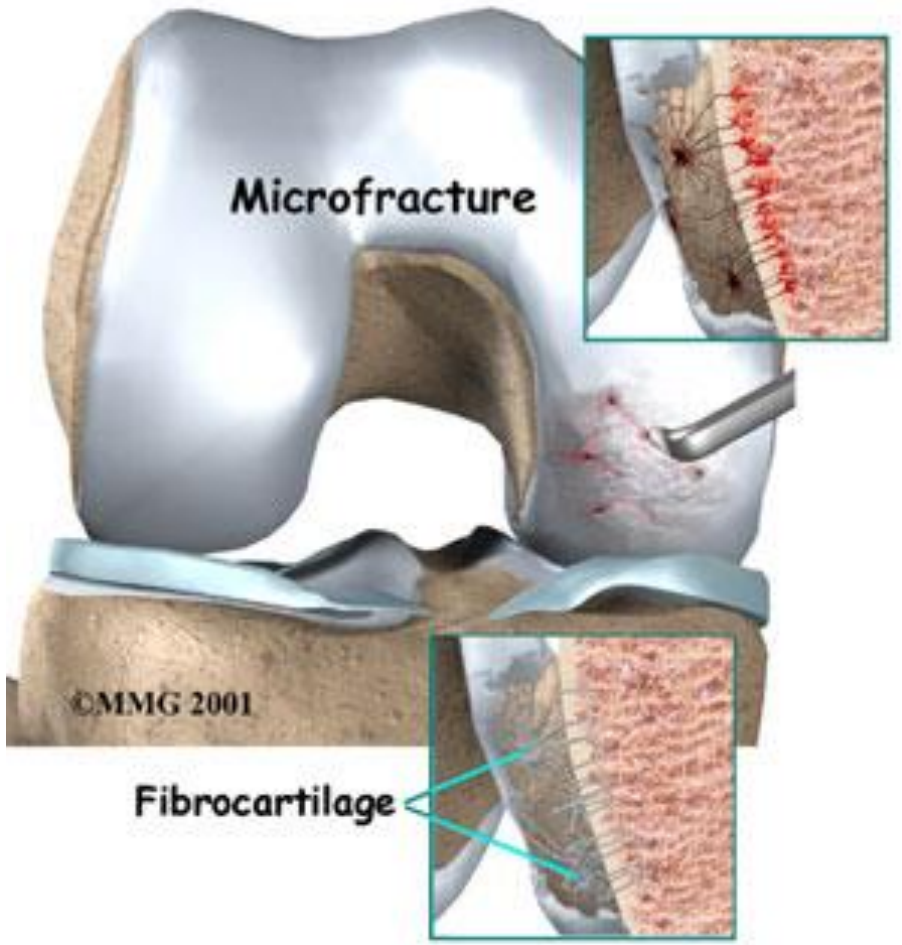
https://www.youtube.com/watch?v=CxhFhidWn6w

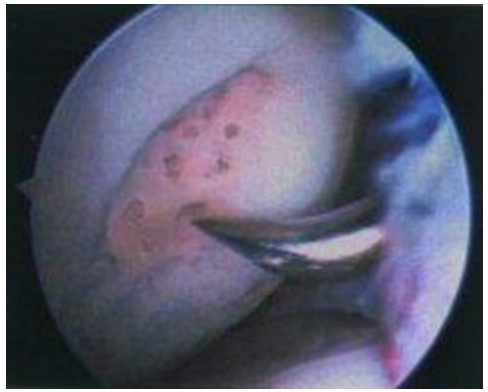
Treatment – Repair

- **Repair** refers to the restoration of a damaged articular surface with new tissue that resembles but does not duplicate the structure, composition and function of articular cartilage

Repair – Microfracture

- Arthroscopic
- Marrow stimulating technique
- Perforations in subchondral bone
- Formation of a ‘super blood clot’
- Fibrocartilage repair material





Repair – Autografts



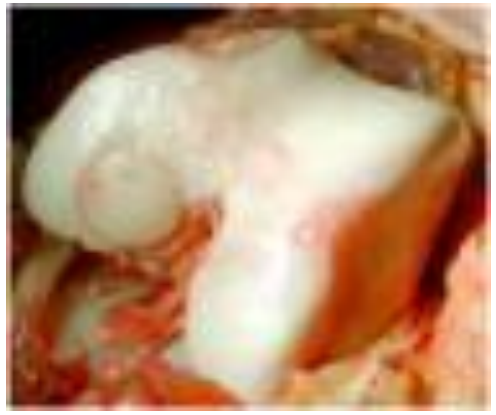
- OATS = Osteochondral Autograft Transfer System
- Plugs taken from non-weight bearing area
- Placement and orientation important
- Small to medium sized defects
- ‘Robbing Peter to pay Paul’



Repair – Allografts

- Allograft osteochondral grafts
- Donor plugs
- No donor site morbidity
- Press-fit technique
- Younger patients
- Aiming to delay premature joint replacement



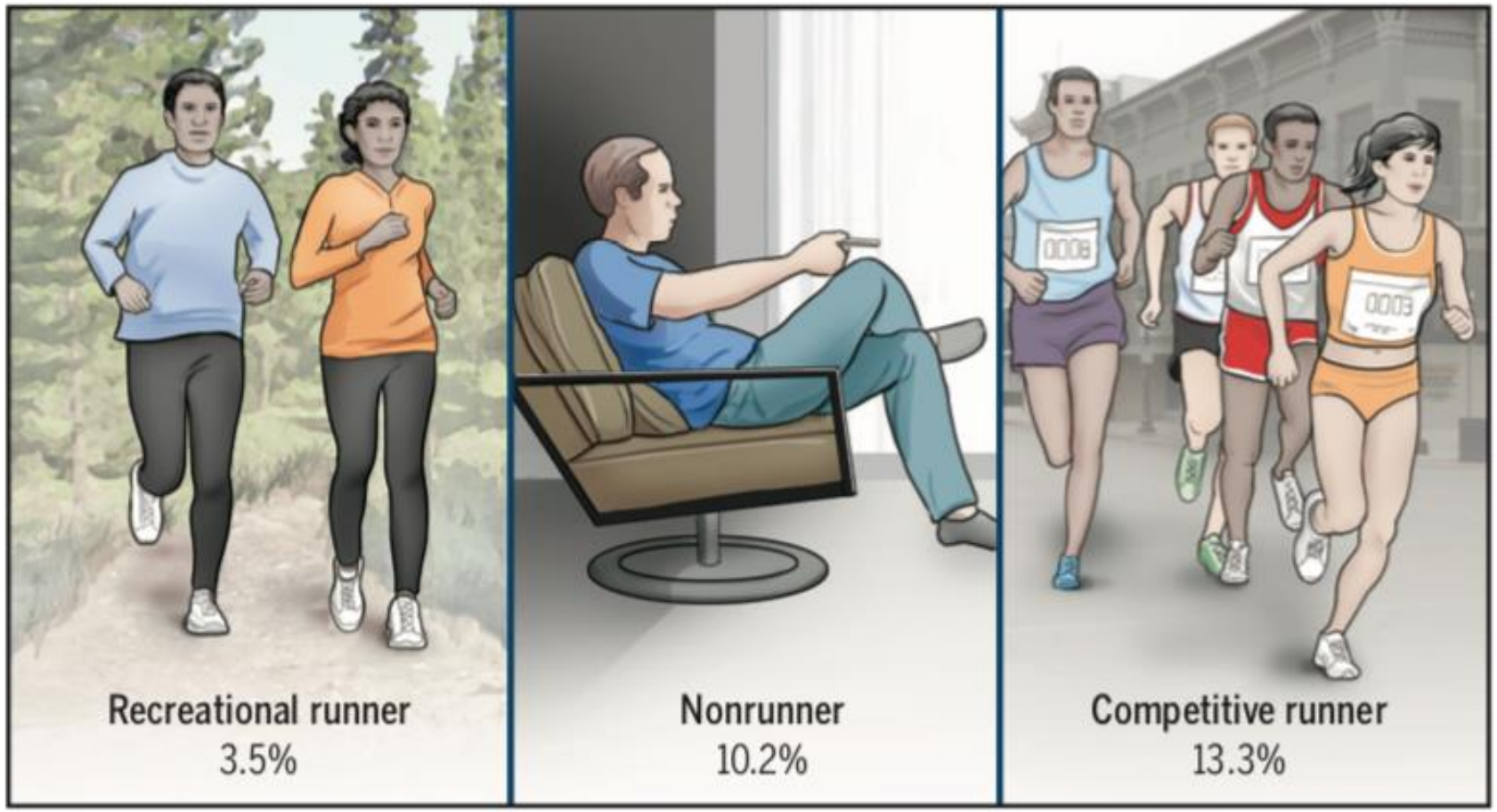


Post Operative Rehabilitation

- Cartilage repair surgery probably has the longest rehabilitation of any elective knee surgery

- Primary rehabilitation goals –
  - Local adaptation & remodelling of the repair
  - Return to function

- Rehabilitation should provide the optimal environment – mechanically & psychologically



BENEFITS OF RECREATIONAL RUNNING. Recreational running is not only good for your overall health, but also benefits your knees and hips—just 3.5% of these runners develop hip or knee arthritis. A sedentary lifestyle—not running—or competing as an elite runner increases the risk of hip or knee arthritis by 10.2% and 13.3%, respectively.

This *JOSPT* Perspectives for Patients is based on an article by Alentorn-Geli et al, titled "The Association of Recreational and Competitive Running With Hip and Knee Osteoarthritis: A Systematic Review and Meta-analysis" (*J Orthop Sports Phys Ther 2017;47(6):373-390. doi:10.2519/jospt.2017.7137*).